G2 operator quick guide

Petoi Bittle X with a Raspberry Pi: safety, stopping, power. Full version: docs/guides/operating-g2.md

Stop G2

Situation	Do this
Anything goes wrong	Say **"emergency stop"** (also "freeze", "halt", "abort"). Motion is cut and he holds. Say **"resume"** to release.
Voice not working	`python -m pi_pipeline.app --halt` on the Pi (release with `--release`).
Moving, nothing in control	`python -m pi_pipeline.link.check_serial send d` on the Pi (rest).
Exploration running	Say "end exploration mode" or run `bash tools/g2_explore.sh stop`.
He fell	The session stops itself. Pick him up and check him before restarting.

Ground rules

- **Never put G2 on his back.** The Pi and the control board are exposed on top.
- Every test starts on his **feet on the floor**, hands near. No testing held in the air.
- Set the floor before a walking run: `g2floor <label>`.
- **Rest or stand him before stopping anything that controls him**: `bash tools/g2_safe_stop.sh voice|explore`.
- **Never deploy to the Pi while he is walking, exploring or being tested.**
- Never send `Xa` or `XAd` to the control board: it silently breaks its voice commands.

Working on him: bench mode

On the calibration stand, or when running calibration or serial checks, start the app with `python -m pi_pipeline.app --bench`. It blocks autonomous movement so nothing competes for the serial link.

Power

Step	What to do
Power on	Switch the Pi's battery (PiSugar) on; the voice service starts by itself. Switch the robot's own pack on. Wait for "G2 online".
Sleep	He sleeps after about 5 minutes at rest. Say "gee two" to wake him.
Shut down	Say a short, clear "shut down". He lies down and the Pi shuts down after 6 seconds (say "cancel" to stop it). **The PiSugar keeps powering the halted Pi: flip its switch by hand.** Switch the robot's pack off separately.
Charging the Pi	Say "you're plugged in" to pause the Pi's battery warning; "you're unplugged" to restart it. Reboot the Pi after charging.

Batteries

- G2 warns by voice when the robot's pack is low or critical, and rests the legs on a critical reading while walking.
- The Pi's battery has no readout. Its warning counts uptime against the measured full-charge runtime.
- Start sessions with both packs full. Stop if the robot's pack passes about 30% resting or the Pi's low alarm sounds.

When something is off

Symptom	First look
No voice reply	`journalctl -u g2-voice -f` on the Pi.
Unexpected ticks or chirps	`python -m pi_pipeline.link.noise_log 60`
Servo loud, stiff or hot	Stop the walk and rest him. See hardware/servo-thermal.md.
After the fact	`python -m pi_pipeline.diag summarize` shows the latest session.
New hardware, first power-on	`python -m pi_pipeline.doctor`, then `python -m pi_pipeline.bringup`.

Source: docs/guides/operating-g2.md. If this sheet and that page disagree, the page is right.